

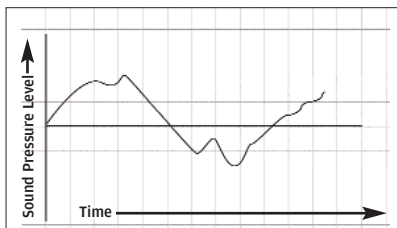
2.1 Sampling Rate, Quantisation, Bitrate

Music (or any sound sequence) is composed of many different frequency signals. A single tone—say the beeps you hear when you get an engaged tone during a phone call—is a single frequency, rather, a non-continuous signal composed of a single frequency. Now, when you listen to one second of music, you're listening to several signals of different frequencies at the same time. Now, music is complex—don't think that a drumbeat (which may sound monotonous by itself) is a single frequency: it's actually composed of signals of many frequencies.

Now think of a signal of a single frequency (of which there are many in one second of music). In the analogue world, this is drawn as a *waveform*, and looks something like this:

Music is composed of a lot of these together. When you look at the spectrum analyser (the light bars moving up and down), remember that what they're showing you is how loud each of the frequencies

is in the music that's being played. When there's a crash of cymbals (lots of treble, or high frequencies), the high frequencies like this one below become more pronounced, and the lights on the right go up.



Now, how is this converted to digital? What we mean by digital is ones and zeroes; so how is a shape like this represented as ones and zeroes? The answer is simple, but read on carefully: what does the waveform

